Cluster Size vs E_T: Data vs MC, Split vs No-Split, Single vs Double Interaction

PPG12 Analysis Team (automated report)

June 2, 2026

1 Motivation

Cluster size (the number of EMCal towers associated with a cluster and the transverse extent of that set of towers) is a basic shape property that underlies every shower-shape-based background discriminant used in PPG12 (NPB, shower-shape BDT, width variables). This report inspects how that size depends on cluster E_T across three orthogonal handles:

- **Sample**: data, signal MC (isolated prompt photons), background MC (QCD jets).
- **Pileup**: single-interaction MC, full-GEANT double-interaction MC, and the physics-weighted mix at 7.9% (1.5 mrad crossing angle) and 22.4% (0 mrad). Only MC is split this way; data is shown as a single curve against the two mixed references.
- **Clusterizer**: the nominal split-cluster container `CLUSTERINFO_CEMC` versus the legacy non-split `CLUSTERINFO_CEMC_NO_SPLIT`. Both containers are stored in every slimtree, so the comparison is an apples-to-apples rerun of the two CEMC clusterizer modes on the same events.

2 Definitions

For each cluster c we read the 7×7 tower matrix $\texttt{cluster\_ownership\_array}_c$ and extract the set of towers marked as owned by the cluster (value 1). Let $\mathcal{O}_c$ be the owned-tower index set, with $|\mathcal{O}_c| = n_{\text{owned}}$. Each index $i \in [0, 48]$ corresponds to a position in the 7×7 grid with $i_\eta = \lfloor i/7 \rfloor$, $i_\phi = i \bmod 7$, the center tower at $(3, 3)$ (see `CaloAna24.cc:1403`).

- n_{owned}: $|\mathcal{O}_c|$ — literal cluster tower count
- Δi_η: $\max_{i\in\mathcal{O}_c} i_\eta - \min_{i\in\mathcal{O}_c} i_\eta + 1$ — bounding-box extent in η
- Δi_ϕ: analogous in ϕ

Both `CLUSTERINFO_CEMC` and `CLUSTERINFO_CEMC_NO_SPLIT` are computed identically. No shower-shape, BDT, NPB, or isolation cut is applied; the only fiducial requirement is $|\eta_{\text{reco}}| < 0.7$.

3 Samples

Category	Samples	Entries (event)	Notes
Data	77 `part_*_with_bdt_split.root`	`DATA_NEVT`	run 47289–54000 (both crossing angle
Signal single	photon5, photon10, photon20	stitched by truth p_T^{γ}	$\sigma_{\text{ref}} = \sigma_{\text{photon20}}$
Signal double	photon10_double	$p_T^{\gamma} \in [10, 100]$	full-GEANT double interaction
Background single	jet5, jet8, jet12, jet20, jet30, jet40	stitched by truth p_T^{jet}	$\sigma_{\text{ref}} = \sigma_{\text{jet50}}$
Background double	jet12_double	$p_T^{\text{jet}} \in [10, 100]$	full-GEANT double interaction

Table 1: Sample inventory. Truth p_T windows (from `CrossSectionWeights.h`) are applied in the analysis macro to avoid double counting across overlapping generator windows.

Per-event cross-section weights are taken from `PPG12::GetSampleConfig` (relative to each class's reference sample). Cluster-level mixing uses a bin-by-bin linear combination of profile means:

$$\langle y \rangle_{\text{mix}}(E_T) = (1 - f)\, \langle y \rangle_{\text{single}}(E_T) + f\, \langle y \rangle_{\text{double}}(E_T), \tag{1}$$

with $f = 0.079$ (1.5 mrad) or $f = 0.224$ (0 mrad). Note that this is a linear combination of the per-bin means, not a weighted merger of underlying fills — the latter would be biased by the relative event counts of the single and double samples.

4 Results

4.1 Nominal data vs inclusive MC

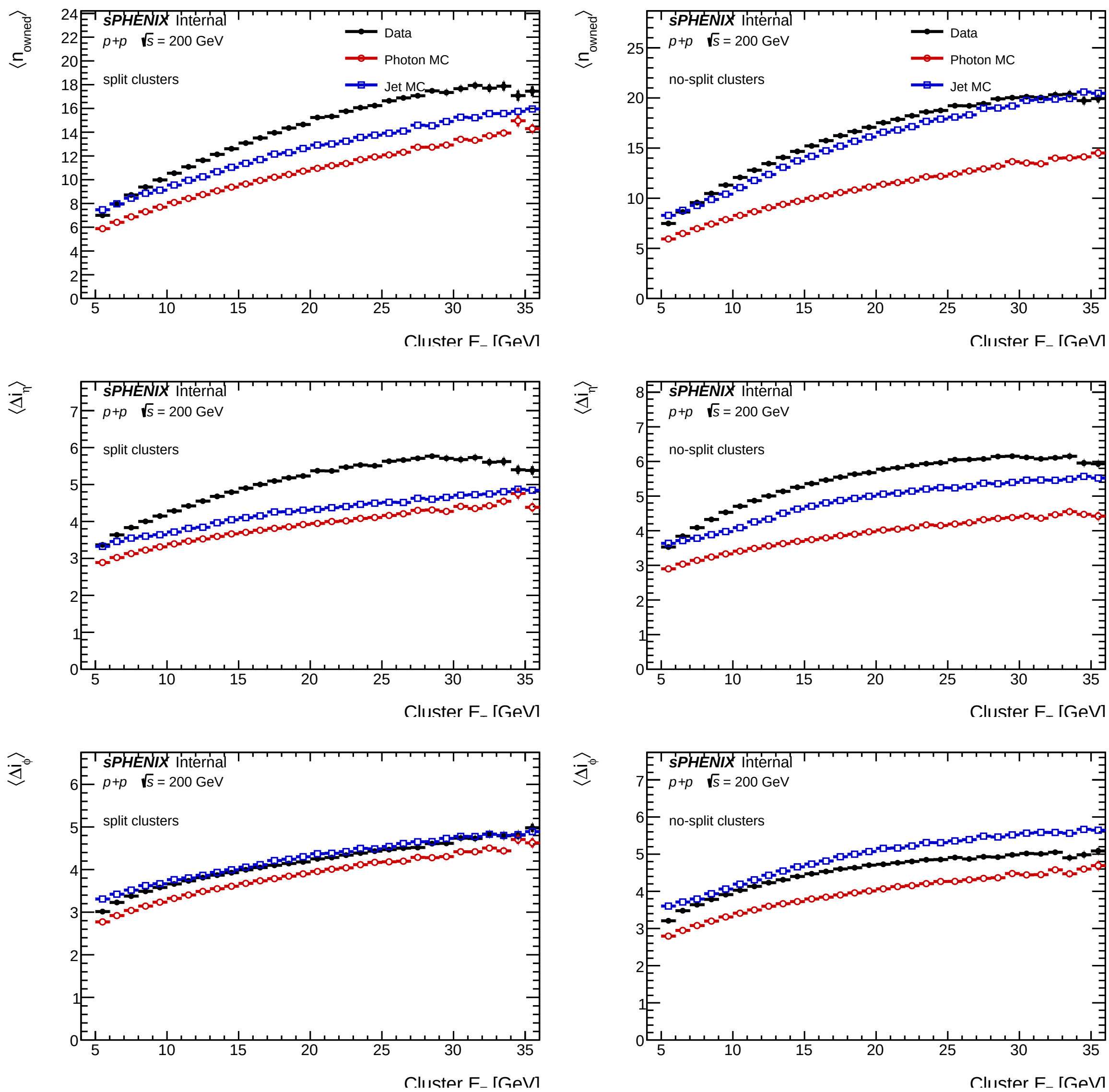

Figure 1: Mean cluster size vs E_T for data (black), pooled signal MC (photon5+10+20, red) and pooled background MC (jet5+8+12+20+30+40, blue), with MC weights from `GetSampleConfig`. Rows: n_{owned}, Δi_η, Δi_ϕ. Columns: split (`CLUSTERINFO_CEMC`), no-split (`CLUSTERINFO_CEMC_NO_SPLIT`). Only $|\eta_{\text{reco}}| < 0.7$ is applied.

Observations.

- **In the split container, data is systematically larger than both MC curves** for n_{owned} and Δi_η. At $E_T = 20\,\text{GeV}$, data has $\langle n_{\text{owned}} \rangle = 15.2$, jet MC has 12.9, photon MC has 11.0 — a data–jet-MC gap of **+2.3 towers** (∼18%). The gap grows from ≈ 0 at $E_T = 5\,\text{GeV}$ to a peak of +2.7 at $E_T = 25\,\text{GeV}$.

- The Δi_ϕ panel is the **exception**: data and jet MC agree within a few percent across the full range. Only the η direction is under-modelled. This asymmetry is consistent with the extra towers being preferentially stretched in η — possibly electronic noise or pedestal drifts on isolated towers that the clusterizer then absorbs into a nearby cluster, plus the larger physical tower pitch in η favouring one-tower spillovers.

- **The data−MC gap is much smaller in the no-split container**: $\approx +1.0$ tower across $E_T = 10$–$25\,\mathrm{GeV}$, shrinking to $+0.4$ at $E_T = 30\,\mathrm{GeV}$ (essentially converged). The extra towers in data *are* picked up by the no-split clusterizer but not fragmented off by the split one — consistent with a diffuse, noise-like contribution spread across the 7×7 matrix rather than a separable sub-cluster. If these were genuine secondary showers, the split algorithm would have removed them, leaving the data−MC gap the same in both containers.
- Jet clusters (blue) are larger than photon clusters (red) across the full E_T range in every metric, as expected — jets deposit energy over multiple overlapping subshowers while prompt photons produce a compact EM shower. The gap widens in `CLUSTERINFO_CEMC_NO_SPLIT` because splitting preferentially shrinks jet clusters while leaving photons alone (quantified in Fig. 4).
- Both MC populations rise with E_T because the $70\,\mathrm{MeV}$ tower threshold becomes easier to exceed as the total shower energy grows. Data shows the same trend with a slightly steeper slope — the data−jet-MC gap in the split container grows from essentially zero at $E_T = 5\,\mathrm{GeV}$ to $\sim$2.7 towers at $E_T = 25\,\mathrm{GeV}$, then starts levelling off.

4.2 Single vs double interaction — photon signal

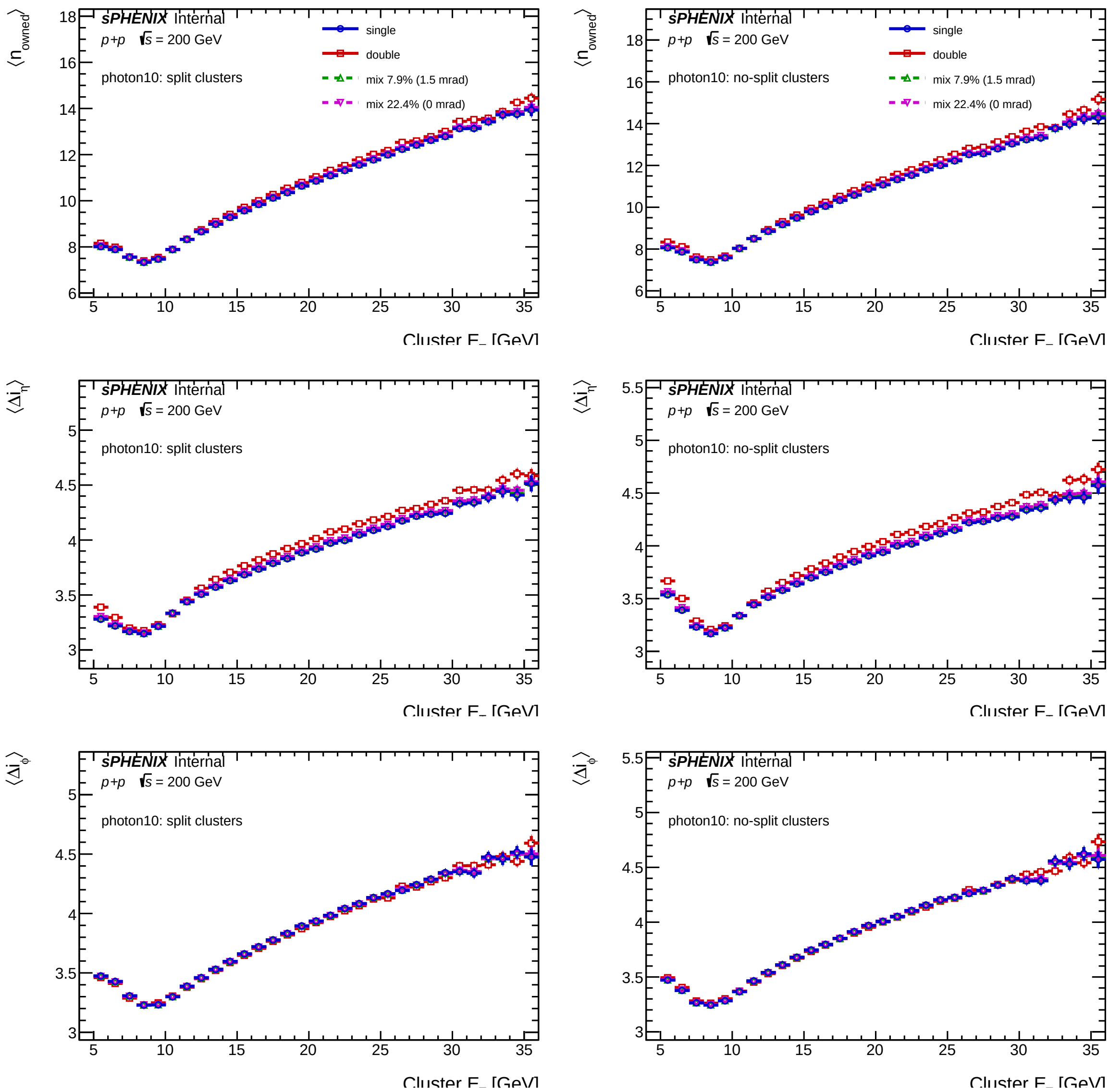

Figure 2: Photon MC cluster size vs E_T: single interaction (`photon10_nom`, blue), pure double interaction (`photon10_double`, red), and linearly mixed at 7.9% (green, 1.5 mrad) and 22.4% (magenta, 0 mrad). Mixing applied per Eq. (1).

Observations.

- Double interaction shifts photon cluster size *upwards* by $\Delta\langle n_{\rm owned}\rangle \approx +0.15$ to $+0.22$ towers above single across $E_T \in [15, 30]$ GeV. The effect is real but small — pile-up adds a few soft neighbours to an otherwise compact EM shower.

- The bounding-box widths (Δi_η, Δi_ϕ) shift by $\lesssim 0.1$ between single and double; the 1st-order effect of double interaction on photon shower width is at the sub-0.1 index level.

- Mixed curves at 7.9% (1.5 mrad) and 22.4% (0 mrad) sit between single and pure-DI at the corresponding fractions. The mixed-7.9% curve is nearly indistinguishable from single — consistent with the shower-shape BDT being pile-up-robust in the 1.5 mrad regime.

- The dip near $E_T = 9$ GeV is the energy-resolution tail of the $p_T^{\gamma,\rm truth} > 10$ generator window; below-threshold reco clusters here are dominated by leakage, not peak showers.

- The split vs no-split containers behave almost identically for photon MC (see Fig. 4), so the DI effect is also essentially independent of the clusterizer.

4.3 Single vs double interaction — jet background

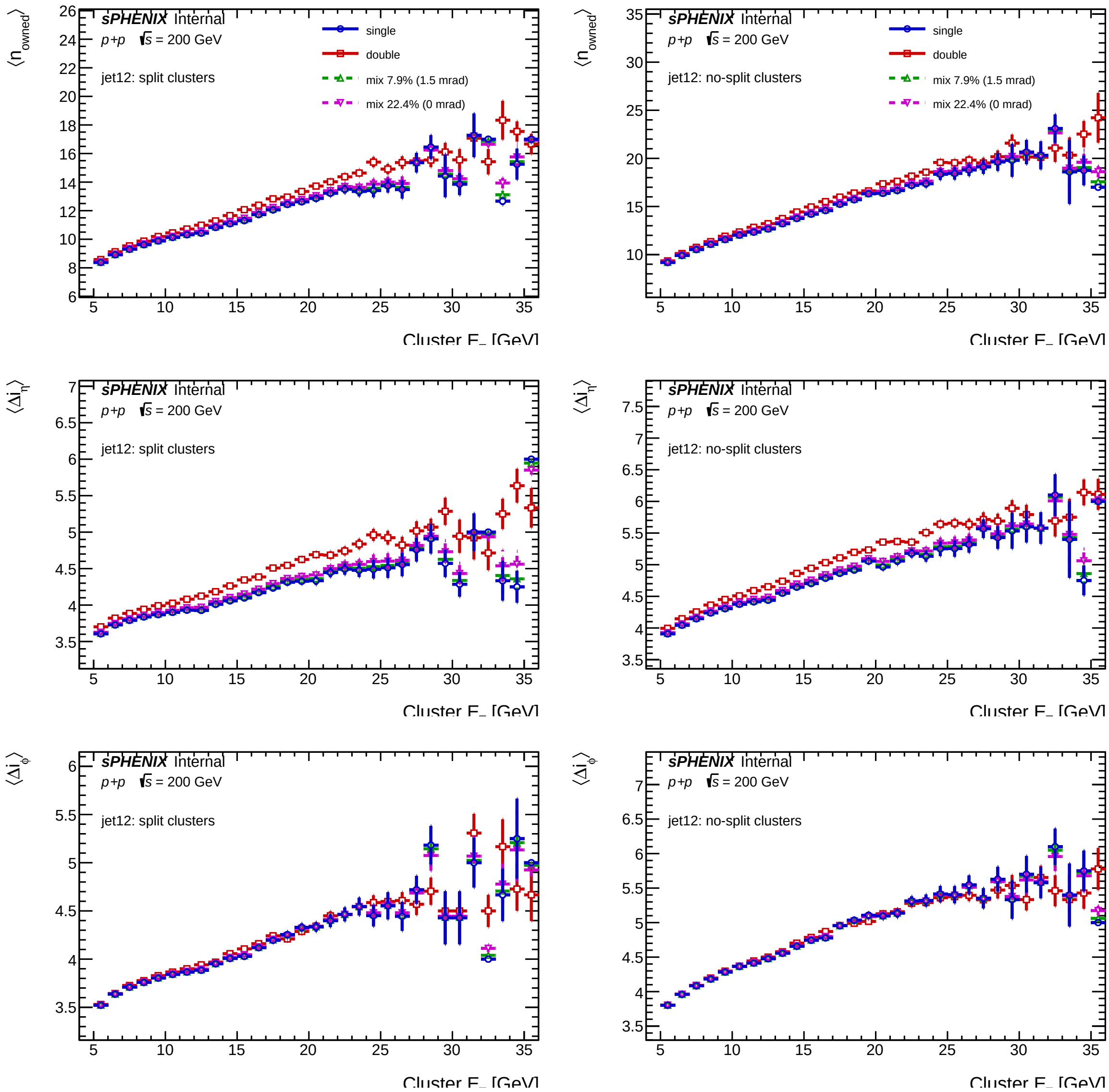

Figure 3: Jet MC cluster size vs E_T, same format as Fig. 2 but for `jet12_nom` vs `jet12_double`.

Observations.

- Double interaction inflates *jet* cluster size much more strongly than photon cluster size: $\Delta\langle n_{\text{owned}}\rangle \approx +0.5$ to $+1.3$ towers in the split container at $E_T > 15\,\text{GeV}$, roughly $4\times$ larger than the photon response.

- Physically: a jet cluster already has multiple soft neighbouring towers from accompanying hadrons, and DI adds more of the same; the probability that DI overlay lands on a low-E tower and pushes it over the 70 MeV threshold is much higher in a jet environment than for a clean photon.

- The DI-induced broadening is partially absorbed by the no-split container (right column), which already inflates jet clusters via overlap, so the relative DI effect is smaller there.

- At $E_T > 30\,\text{GeV}$ the `jet12` sample runs out of statistics (truth $p_T^{\text{jet}} \in [14, 21]\,\text{GeV}$ window), producing the noisy tail seen in every jet12 DI panel.
- Because jets enter both the fake-photon and anti-tight control regions of the ABCD, this DI effect is a non-trivial input to the background-template closure: in 0 mrad running (22.4% DI), the jet template mean cluster size shifts upward by ∼0.3–0.4 towers, which affects the shower-shape BDT input distribution and therefore the background rejection rate.

4.4 No-split / split ratio

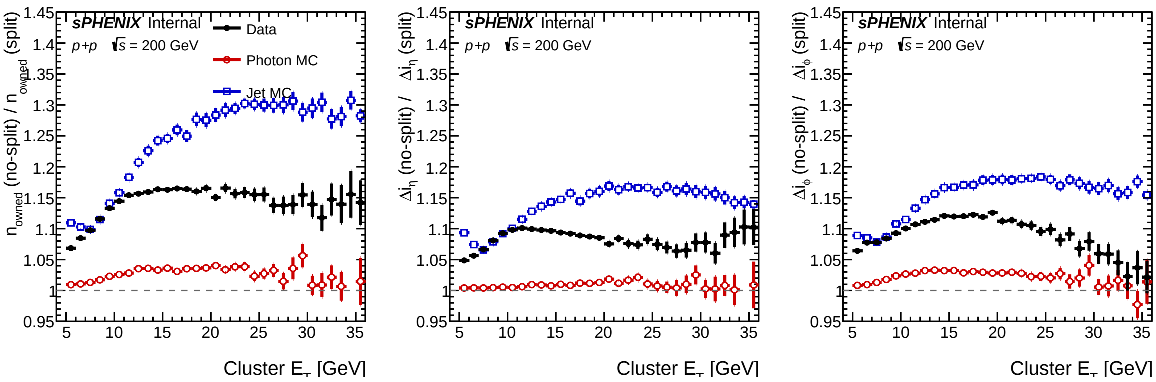

Figure 4: Ratio of mean cluster size in the no-split container to that in the split container, vs E_T. Solid black: data. Red: pooled photon MC. Blue: pooled jet MC. A flat ratio of 1 would mean the two clusterizer modes are indistinguishable.

Observations.

- **Photon MC (red)** has a nearly flat ratio of 1.01–1.04 in every panel: splitting has essentially no effect on clean single-photon clusters.
- **Jet MC (blue)** shows a clear E_T-dependent effect: n_{owned} ratio rises from ≈1.10 at $E_T = 7\,\text{GeV}$ to ≈1.30 at $E_T \gtrsim 20\,\text{GeV}$. The bounding-box widths (Δi_η, Δi_ϕ) show the same trend at a milder level, saturating around 1.16–1.18. Splitting removes ∼30% of jet cluster towers at $E_T > 15\,\text{GeV}$.
- **Data (black) sits between photon MC and jet MC**, with a split/no-split ratio of ≈1.10–1.17 for n_{owned} — below the jet-MC expectation of 1.30. This means the extra towers that data has compared to MC (Sec. 4.1) are *not* being removed by the split clusterizer — they are distributed diffusely, not as separable sub-clumps. Physically consistent with noise/threshold contamination rather than genuine secondary showers.
- The data n_{owned} ratio peaks near $E_T = 15$–$20\,\text{GeV}$ and declines slightly at higher E_T, whereas jet MC stays flat or gently rises. Data is approaching the 49-tower matrix cap in `CLUSTERINFO_CEMC_NO_SPLIT` at $E_T \gtrsim 30\,\text{GeV}$, so the ratio numerator saturates before the denominator.

5 Summary

Four findings matter for the PPG12 analysis:

1. **Data is systematically larger than jet MC in the split container** in $\langle n_{\text{owned}} \rangle$ and $\langle \Delta i_\eta \rangle$ by +1 to +2.7 towers (∼10–20% of the mean) across $E_T = 10$–$30\,\text{GeV}$. The data−MC gap is ∼3× smaller in the no-split container. $\langle \Delta i_\phi \rangle$ data-MC agreement is *much* better. This is the

single biggest finding: the clusterizer MC does not reproduce the tower-multiplicity of real data. Most plausible causes are noise-threshold miscalibration or un-modelled underlying-event / zero-bias contamination; electronic noise is the likeliest root cause given the η-only asymmetry and the absence of a matching shift in ϕ.

2. **The extra data-MC towers are diffuse, not clumped**. In the no-split/split ratio (Fig. 4), data sits *below* jet MC: extra towers in data do not form separable sub-clusters that the split algorithm could fragment off, so splitting has less to remove. The implication for the PPG12 shower-shape BDT is that it should be trained and validated against data-like (not MC-like) cluster morphology to avoid over-tightening at high E_T.

3. **Cluster splitting is a photon-safe jet shrinker**. The split clusterizer removes ∼30% of jet cluster towers at $E_T > 15\,\mathrm{GeV}$ while leaving photon cluster size unchanged to within a few percent (red curve in Fig. 4). This is the physical mechanism by which `CLUSTERINFO_CEMC` improves signal-background shape separation and motivates the nominal pipeline's choice of the split container.

4. **Double interaction broadens jet clusters much more than photon clusters**. Pure DI adds ∼0.5–1.3 towers to jet cluster size (4–10% of the mean) and only ∼0.2 towers to photon cluster size (2% of the mean). At 1.5 mrad running (7.9% DI), the effect on the weighted mean is absorbed by systematics, but at 0 mrad (22.4% DI) the jet-cluster shift is a concrete mechanism for the DI systematic on the background template shape. Photon DI is sub-leading.

Follow-ups suggested by this study

- **Investigate the data-MC cluster-size gap**. Cross-check by running the same macro on a `minbias`-triggered data subset, with and without a $|t_{\mathrm{MBD}}| < 25\,\mathrm{ns}$ cut, to isolate whether the gap is from noise, underlying event, or pile-up.
- Compare the noise-tower distributions between data and MC (per-channel RMS, hit rates above the 70 MeV threshold in empty fiducial regions) to quantify the noise-contamination hypothesis.
- Once the noise/threshold question is settled, decide whether the BDT training sample should include a data-like noise overlay.

Reproducibility

Generation:

```
cd efficiencytool && bash run_cluster_size.sh
cd ../plotting && root -l -b -q plot_cluster_size.C+
```

Key macros:

- `efficiencytool/ClusterSizeStudy.C` — computes n_{owned}, Δi_η, Δi_ϕ per cluster, both cluster nodes, with truth-p_T sample stitching and fiducial $|\eta_{\mathrm{reco}}| < 0.7$.
- `efficiencytool/run_cluster_size.sh` — driver (6-way parallel).
- `plotting/plot_cluster_size.C` — figures.